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ANKLE JOINT (TIBIO-TARSIAL)

PLAN

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Objectives:

- To be able to classify a joint.
- To know the articular surfaces and their means of connection.
- To know the movements and axes of a joint.

ANKLE JOINT

I. INTRODUCTION:

The ankle joint, or tibiotalar joint, is a type of ginglymic (trochlear) diarthrosis connecting the two bones of the lower leg (tibia and fibula) to the talus. It bears the entire weight of the body and plays a crucial role in foot stability and the gait cycle. It is the joint most prone to sprains.

II. DESCRIPTIVE ANATOMY:

1. **ARTICULAR SURFACES:** Covered with cartilage, these are the articular surfaces of the tibia and fibula, which form a tibiofibular mortise into which the talar tenon is fitted.

A. **The tibiofibular mortise:** It is formed by the distal ends of the tibia and fibula and comprises a roof and two lateral walls.
a) The roof (superior wall): This is the inferior surface of the tibial plafond, quadrilateral in shape, elongated transversely, and concave from front to back. It is divided into two surfaces by a rounded anteroposterior ridge (crest) related to the groove of the talar pulley.

b) The medial wall: This is the lateral surface of the medial malleolus of the tibia. It is flat, vertical, and triangular with its base anterior, continuous with the superior wall, and it articulates with the malleolar surface of the talus.

c) The lateral wall: This is the articular surface of the medial aspect of the lateral malleolus, triangular in shape with its base superior and its apex inferior, and it articulates with the lateral surface of the talus.

B. The talar tenon:

It fits into the mortise and is formed by the superior, medial, and lateral surfaces of the talus, all covered with cartilage.

a) The superior surface: This is the trochlea of the talus (talar pulley). It has a medial facet, a lateral facet, and a groove. The lateral facet is wider and higher than the medial one. It articulates with the tibial plafond.

b) The medial surface (medial malleolar surface): This is comma-shaped with a large anterior end and articulates with the medial malleolar surface.

c) The lateral surface (lateral malleolar surface):

It is a triangle with a superior base, corresponding to the lateral malleolus.

2. MEANS OF UNION:

A. The articular capsule: A fibrous sleeve that attaches to the articular surfaces. It is strong laterally, reinforced by the lateral ligaments, and very loose anteriorly and posteriorly, reinforced by fibrous tracts forming the anterior and posterior ligaments.

B. Passive ligaments: There are four of them: the thin anterior and posterior ligaments, and especially the collateral ligaments.

a) **Tibial collateral ligament:** very strong, it is made up of two fibrous layers:

- Superficial layer: **Farabeuf's deltoid ligament:**

  Originating from the anterior border of the tibial malleolus, it runs downwards and forwards, widening in a fan shape, and inserts on:

- The superior surface of the navicular bone (scaphoid)

- The medial border of the plantar calcaneonavicular ligament and on the sustentaculum tali of the calcaneus.

- Deep layer: formed of two bundles:
  - **Anterior bundle (anterior tibiotalar ligament):**

Extends from the anterior border of the medial malleolus to the medial aspect of the talar neck.

- **Posterior bundle (posterior tibiotalar ligament):** Extends from the apex of the medial malleolus to the medial surface of the talus.

b) **Fibular collateral ligament:** formed of three bundles:

- ✓ **Anterior bundle: anterior talofibular ligament:** extends from the lateral malleolus to the lateral surface of the talar neck.
- ✓ **Middle bundle: calcaneofibular ligament:** extends from the lateral malleolus to the lateral surface of the calcaneus.
- ✓ **Posterior bundle: posterior talofibular ligament:** extends from the lateral malleolus to the posterior surface of the talus.

c) Calcaneotalofibular ligament: Inconstant, it extends from the posterior border of the lateral malleolus to the posterior surface of the talus and the superior surface of the calcaneus.

d) Anterior ligament: **Anterior tibiotalar ligament:** Very weak, it is composed of oblique fibers extending from the anterior border of the tibia to the lateral surface of the neck of the talus.

e) Posterior ligament: Two in number:

- **Posterior tibiofibular ligament:** Extends horizontally between the medial and lateral malleoli.
- **Calcaneotalofibular ligament:** Extends between the lateral malleolus and the posterior tarsal bones.

C. Active ligaments: The Achilles tendon, the peroneus brevis and longus muscles, and the flexor and extensor muscles

II. MEANS OF GLIDING:

The synovium lines the deep surface of the capsule, reflects at its bony insertion, and continues to the edge of the cartilage. It forms a superior extension, creating a cul-de-sac between the tibia and the fibula.

IV. FUNCTIONAL ANATOMY:

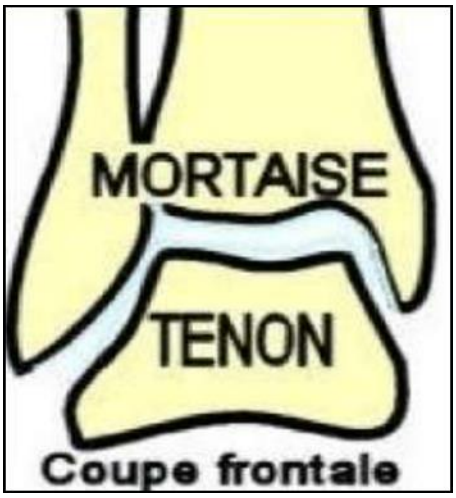
This is a joint with a single degree of freedom, allowing only flexion-extension movements.

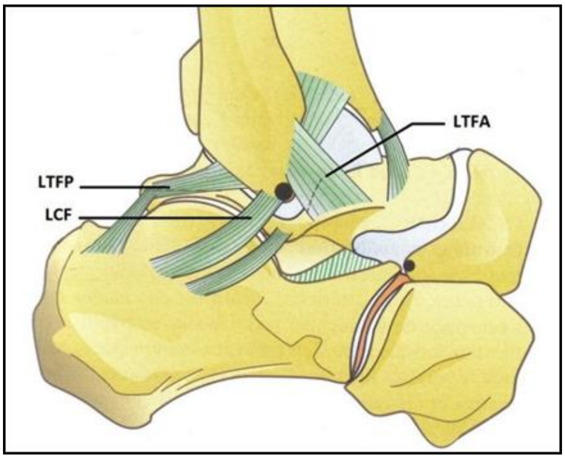
The axis of movement is transverse:

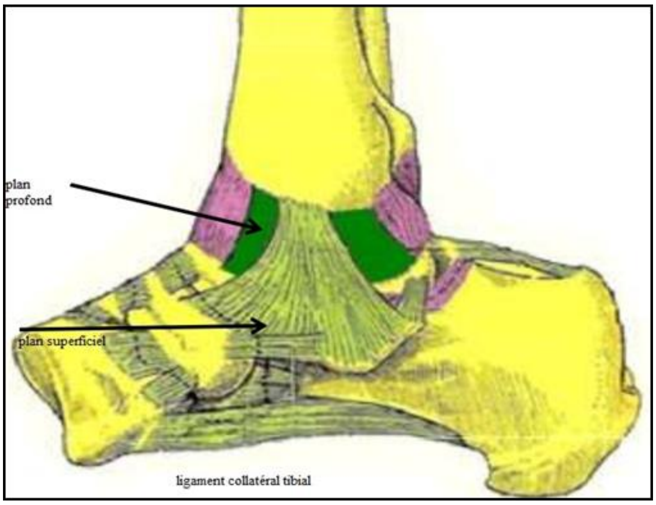
- **Dorsiflexion of the ankle:** a movement that attempts to bring the dorsal surface of the foot towards the anterior surface of the leg. Range of motion is 20°.
- **Plantar flexion:** a movement that moves the dorsal surface of the foot away from the anterior surface of the leg. Range of motion is 30°-40°.

V. CONCLUSION:

The ankle joint is a ginglyme type diarthrosis, allowing only flexion-extension movements because it is laterally restricted by the malleoli and strong collateral ligaments. It is examined by: standard X-ray, CT scan, MRI, arthroscopy, and arthrography. It is frequently prone to sprains.


MORTAISE
TENON
Coupe frontale


LTFA
LTFP
LCF


plan profond
plan superficiel
ligament collatéral tibial

FOOT JOINTS

PLAN :

I. **INTRODUCTION**
II. **DESCRIPTIVE ANATOMY**
  A. **TARSAL JOINTS**
  B. **TARSOMETATARSAL JOINTS**
  C. **INTERMETATARSAL JOINTS**
  D. **METATARSOPHALANGIAL JOINTS**
  E. **INTERPHALANGIAL JOINTS**
III. **FUNCTIONAL ANATOMY**

I. INTRODUCTION:

The joints of the foot are complex and numerous, and are divided into 5 groups:

- Tarsal joints

- Tarsometatarsal joints

- Intermetatarsal joints

- Metatarsophalangeal joints

- Interphalangeal joints.

II. **DESCRIPTIVE ANATOMY:**

A. TARSAL JOINTS:

1. Posterior Tarsal Joint: Subtalar:

Unites the talus and the calcaneus; it is a double trochoid diarthrosis.

➢ The articular surfaces are:

- The posterior calcaneal surface of the talus: located on the inferior surface of the body of the talus, it is concave.

- The posterior talar surface of the calcaneus: it is convex, located on the middle part of the posterior surface of the calcaneus.

➢ Means of union: a capsule and the ligaments:

- Lateral talocalcaneal ligament.

- Medial talocalcaneal ligament.

- Posterior talocalcaneal ligament.

- Interosseous talocalcaneal ligament.

2. Midtarsal joint (CHOPART);

Unites the posterior and anterior tarsi. It has an elongated S-shape and is composed of two articulations:

- Calcaneocuboid joint: a saddle-shaped synovial joint.

- Talocalcaneonavicular joint: a ball-and-socket synovial joint.

3. Joints of the anterior tarsus:

The bones of the anterior tarsus are joined by five articulations:

- Cuneonavicular joint

- Cuboidonavicular joint

- Medial and lateral intercuneiform joints

- Cuneocuboid joint.

B. TARSOMETATARSAL JOINTS (LISFRANC): These joints unite the three cuneiform bones and the cuboid with the five metatarsals. It is a diarthrosis comprising several joints. It is divided into three distinct articulations:

- The medial joint: unites the medial cuneiform with the first metatarsal.

- The middle joint: unites the intermediate and lateral cuneiforms with the second and third metatarsals.

- The lateral joint: unites the cuboid with the fourth and fifth metatarsals.

➢ The articular surfaces are represented by:

- The tarsal arch, which is formed by the articular surfaces of the three cuneiforms and the articular surface of the cuboid.

- The metatarsal arch, which is formed by the articular surfaces of the 1st, 2nd, 3rd, 4th, and 5th metatarsals.

➢ The means of union are represented by capsules reinforced by three types of ligaments:

- Dorsal tarsometatarsal ligaments

- Plantar tarsometatarsal ligaments

- Interosseous cuneometatarsal ligaments

C. INTERMETATARSAL JOINTS: These joints unite the bases of the metatarsals and are diarthroses of the arthrodial type.

- The base of the first metatarsal does not articulate with the base of the second, but they are united only by a few fibrous bundles.

- The articular surfaces are represented by the lateral faces of the bases of the second, third, fourth, and fifth metatarsals.

- Each intermetatarsal joint has a capsule reinforced by three metatarsal ligaments: dorsal, plantar, and interosseous.

D. METATARSOPHALANGEAL JOINTS:

These are ball-and-socket joints that connect the heads of the metatarsals to the bases of the proximal phalanges.

➢ The articular surfaces are represented by:

- The articular surfaces of the metatarsal heads.

- The glenoid fossae of the proximal phalanges.

- The glenoid fibrocartilage.

- <u>Means of union:</u>

Each joint comprises a loose joint capsule reinforced by ligaments:

- The medial and lateral collateral ligaments

- The plantar ligament

- The transverse ligament

E. INTERPHALANGIAL JOINTS:

These joints connect the phalanges: the 1st to the 2nd and the 2nd to the 3rd. There is a single interphalangeal joint for the big toe, and two for each of the four remaining toes.

- <u>The articular surfaces</u> are represented by:

- The articular surfaces of the base and head of the phalanx.

- The glenoid fibrocartilage.

- <u>Means of union:</u>

Each joint comprises a capsule reinforced by ligaments:

- The medial and lateral collateral ligaments

- The plantar ligament.

III. FUNCTIONAL ANATOMY:

These joints allow two types of movement:

- **Varus or inversion**: which is extension, supination, and adduction (the sole of the foot turns inward).
- **Valgus or eversion:** which is a dorsiflexion, pronation and abduction (the sole of the foot faces outwards).

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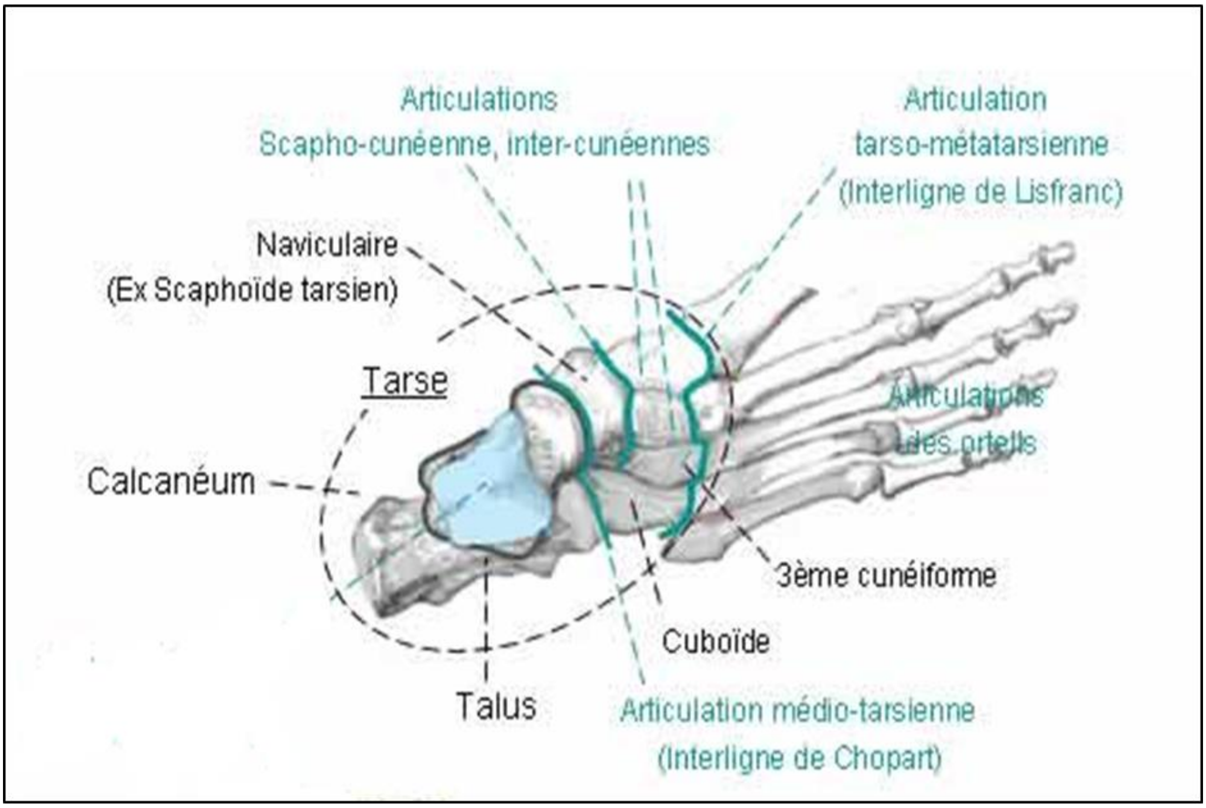
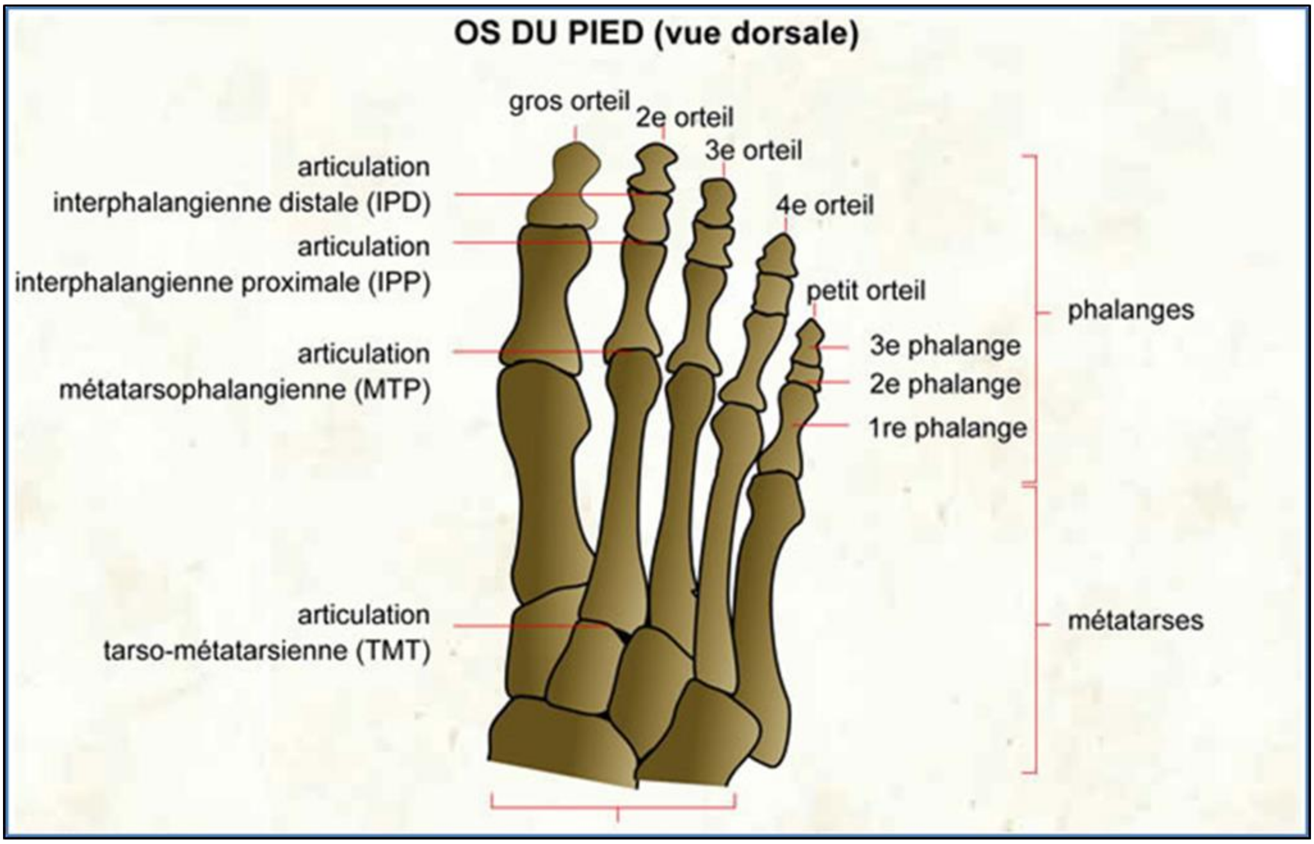

Articulations
Scapho-cunéenne, inter-cunéennes
Articulation
tarso-métatarsienne
(Interligne de Lisfranc)
Naviculaire
(Ex Scaphoïde tarsien)
Tarse
Calcanéum
Articulations
des orteils
3ème cunéiforme
Cuboïde
Talus
Articulation médio-tarsienne
(Interligne de Chopart)


OS DU PIED (vue dorsale)
gros orteil
2e orteil
3e orteil
4e orteil
petit orteil
articulation
interphalangienne distale (IPD)
articulation
interphalangienne proximale (IPP)
articulation
métatarsophalangienne (MTP)
articulation
tarso-métatarsienne (TMT)
3e phalange
2e phalange
1re phalange
phalanges
métatarses